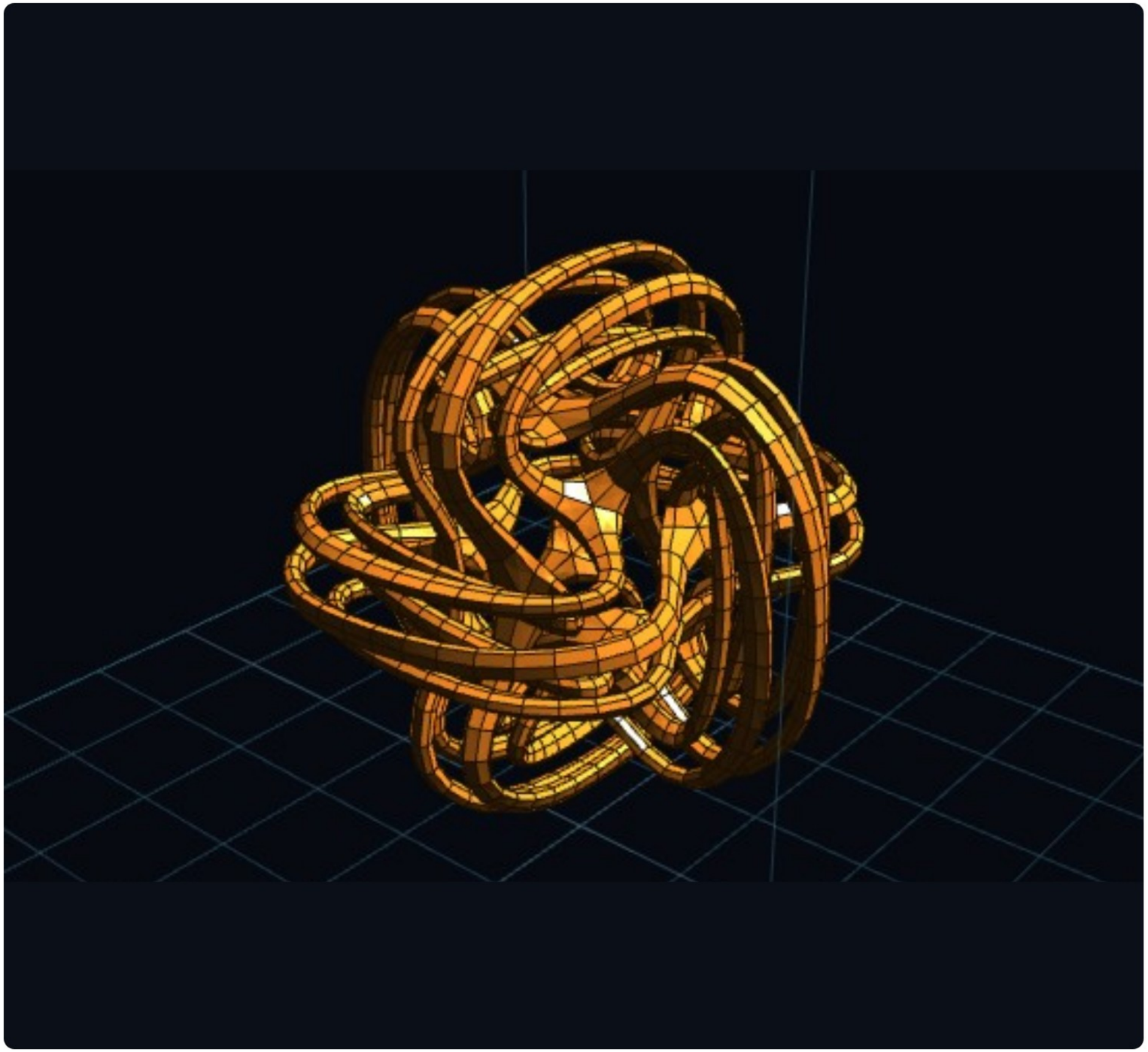


The TopMod Look

Dodecahedron → 6 twisted handles → Doo-Sabin → face loops → rind → Catmull-Clark · step-by-step guide

CREDIT

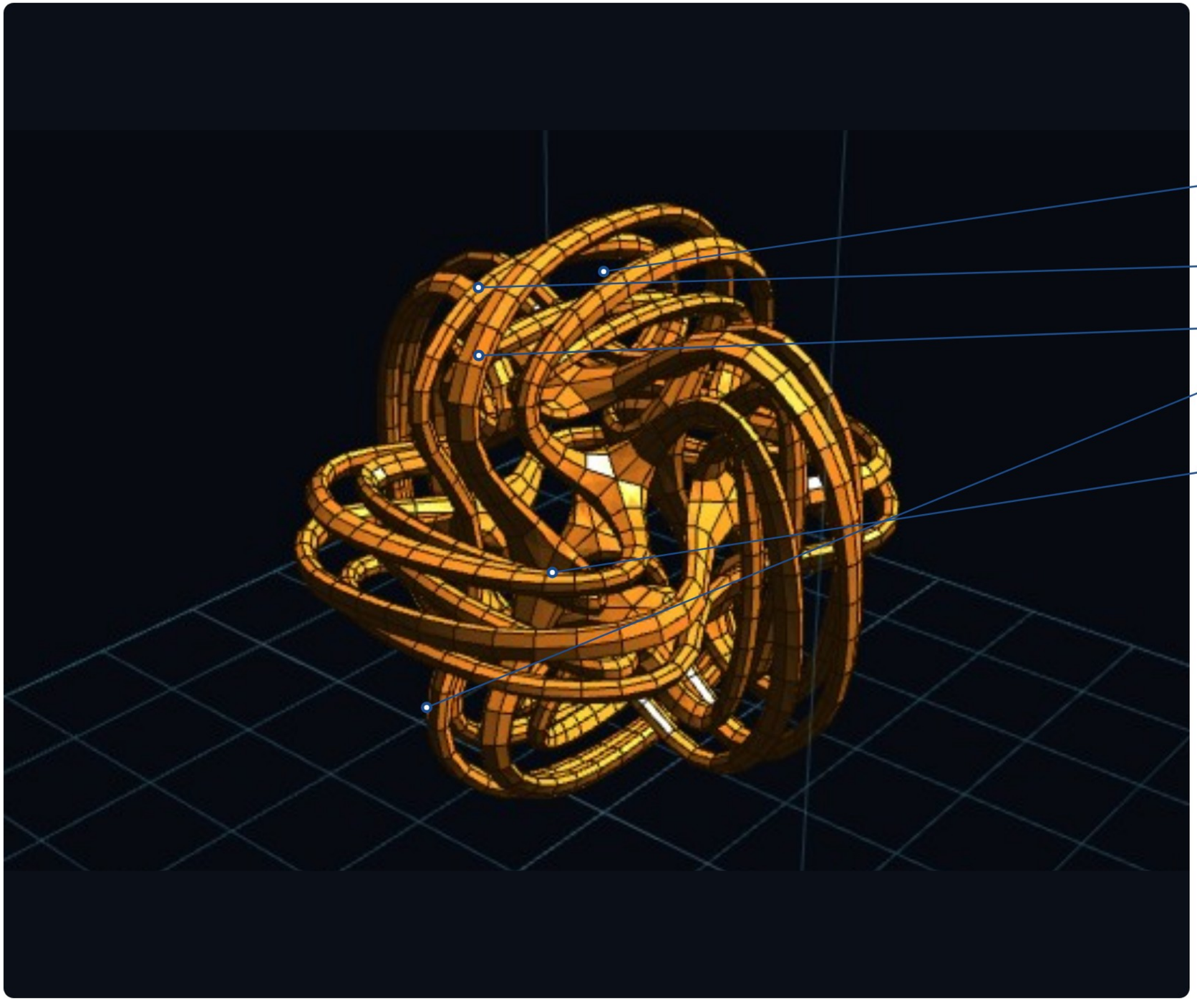
After Nathan Ho, “TopMod and Blender”. TopMod: Ergun Akleman, Vinod Srinivasan, Jianer Chen (Texas A&M).

STEPS

01 Dodecahedron
02 Handle 1 · 72° twist
03 Handles 2–3
04 All 6 handles
05 Doo-Sabin ×1
06 Select face loops
07 Rind modeling
08 Catmull-Clark ×2
09 Undo one level
10 Line Art
11 Sapphire view

Shortcut: Tools ▸ The TopMod Look builds the finished form in one click; File ▸ Export Step Guide compiles your own steps the same way.

PROJECT	SHEET TITLE	SOURCE	DRAWN BY	DATE	STEPS	SHEET
The TopMod Look	**COVER**	TOPMOD 3D v3.9	TOPMOD 3D	2026-10-13	11	1 OF 6 **01**



Anatomy of the form

Catmull-Clark ×1 after rind modeling. Euler–Poincaré: V − E + F = 2 − 2g.

Handle (×6)

Curved tube between two faces; each one adds a hole — genus 6 after all six.

72° twist

The tube's end is offset one corner (360°/5) — the loops spiral.

Slot (removed face loop)

A loop of quads cut out by rind modeling; it weaves from handle to handle.

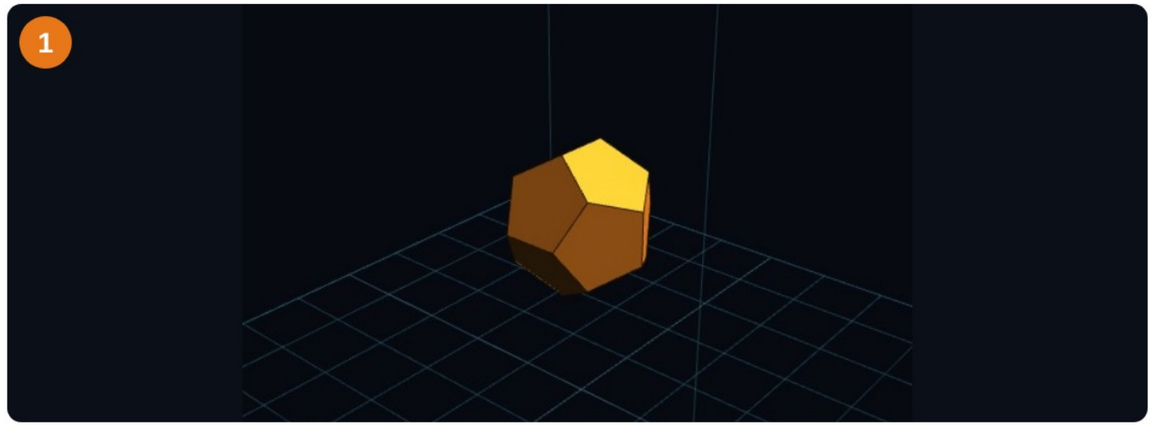
Rind wall

Thickness of the crust: inner shell offset along the vertex normals, walls along the slot outline.

Junction

Where three handles meet — an original dodecahedron vertex (valence 3 → 6).

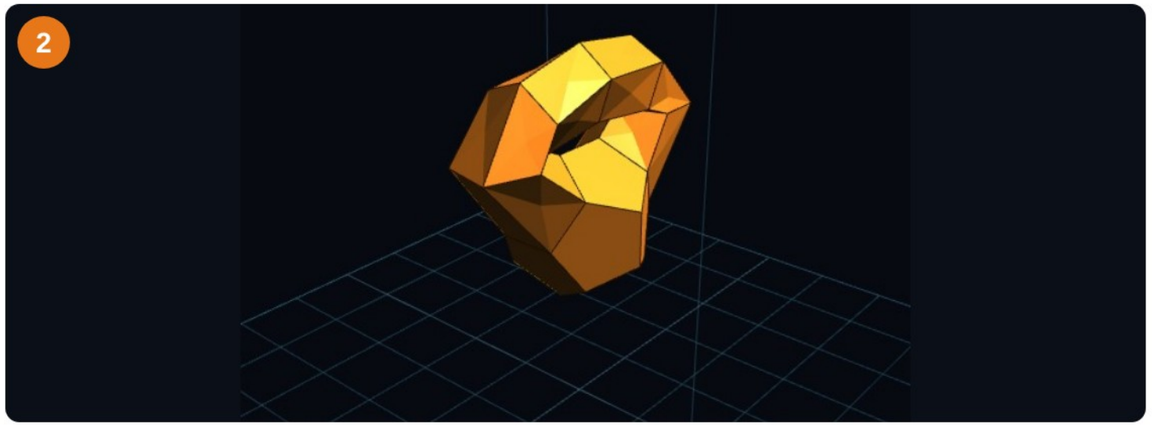
PROJECT	SHEET TITLE	SOURCE	DRAWN BY	DATE	STEPS	SHEET
The TopMod Look	**ANATOMY**	TOPMOD 3D v3.9	TOPMOD 3D	2026-10-13	11	2 OF 6 **02**



Dodecahedron

Tools ▸ Primitives ▸ Dodeca. Twelve pentagons, genus 0 — every face will become the end of a handle.

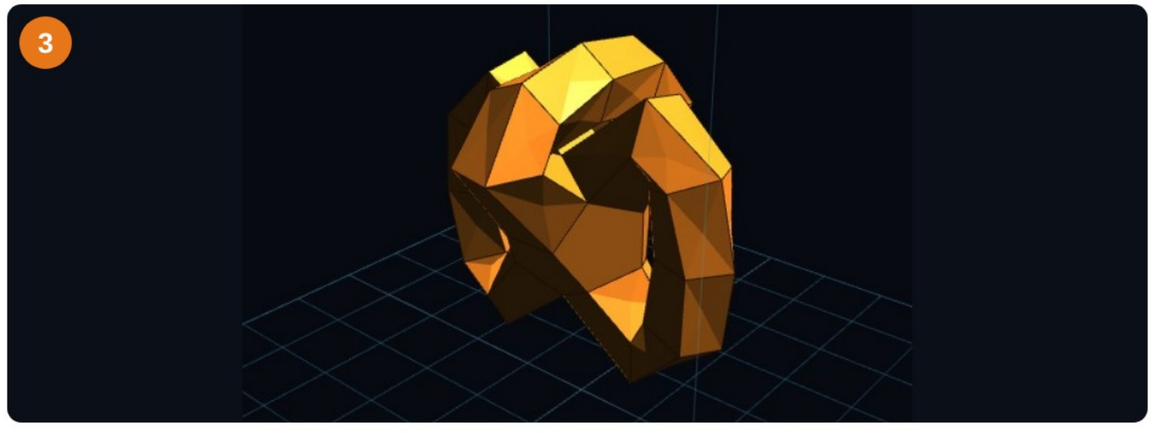
PLATONIC SOLID · DODECAHEDRON
12 pentagons · 30 edges · 20 vertices
{p,q} = {5,3} · dihedral angle = 2·atan(φ) ≈ 116.57°
V 20 − E 30 + F 12 = χ 2 ⇒ genus 0



Handle 1 · 72° twist

Faces mode (3). Pick two faces that share no vertex, then Edit ▸ Add Handle. Click a corner on face A and the next corner round on face B — one step = 72° twist.

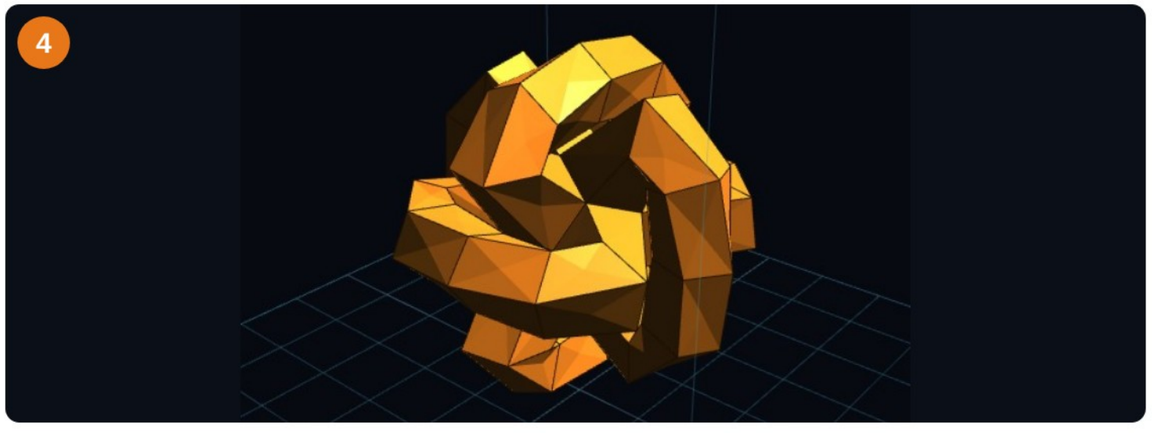
HANDLE (CURVED, TWISTED)
each handle: χ → χ − 2 · g → g + 1
twist θ = k · 360°/n (n = 5 → 72°)
path P(t) = Hermite(c_A, h·n̂_A, c_B, −h·n̂_B)
V 45 − E 85 + F 40 = χ 0 ⇒ genus 1



Handles 2–3

Repeat on the next two face pairs. Each handle adds one to the genus (χ drops by 2).

HANDLE (CURVED, TWISTED)
each handle: χ → χ − 2 · g → g + 1
twist θ = k · 360°/n (n = 5 → 72°)
path P(t) = Hermite(c_A, h·n̂_A, c_B, −h·n̂_B)
V 95 − E 195 + F 96 = χ -4 ⇒ genus 3

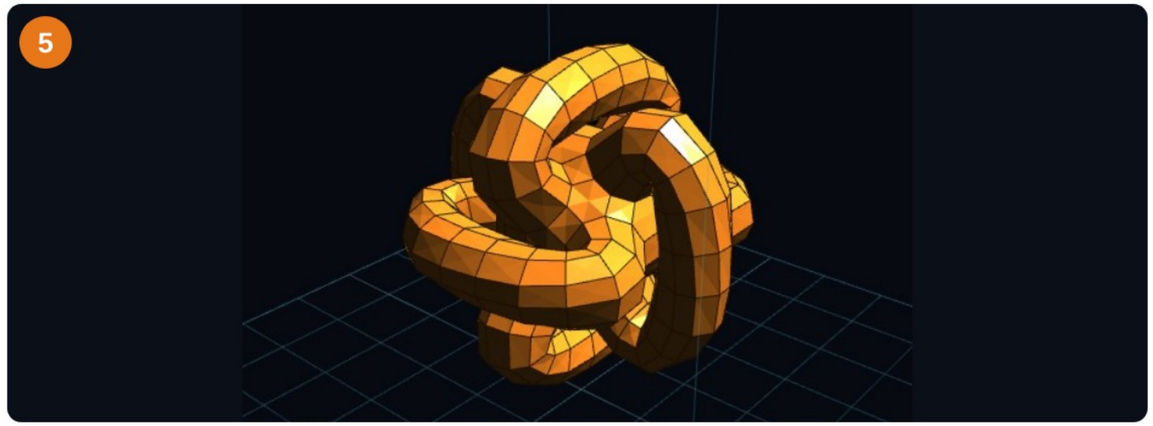


All 6 handles

Six handles use all twelve faces. The pairs are chosen so the loops are evenly spread around the form. Genus 6.

HANDLE (CURVED, TWISTED)
each handle: χ → χ − 2 · g → g + 1
twist θ = k · 360°/n (n = 5 → 72°)
path P(t) = Hermite(c_A, h·n̂_A, c_B, −h·n̂_B)
V 170 − E 360 + F 180 = χ -10 ⇒ genus 6

PROJECT	SHEET TITLE	SOURCE	DRAWN BY	DATE	STEPS	SHEET
The TopMod Look	**STEPS 1–4**	TOPMOD 3D v3.9	TOPMOD 3D	2026-10-13	11	3 OF 6 **03**



Doo-Sabin ×1

Remeshing ▸ Doo-Sabin, once. This corner-cutting pass is what gives the TopMod look its texture.

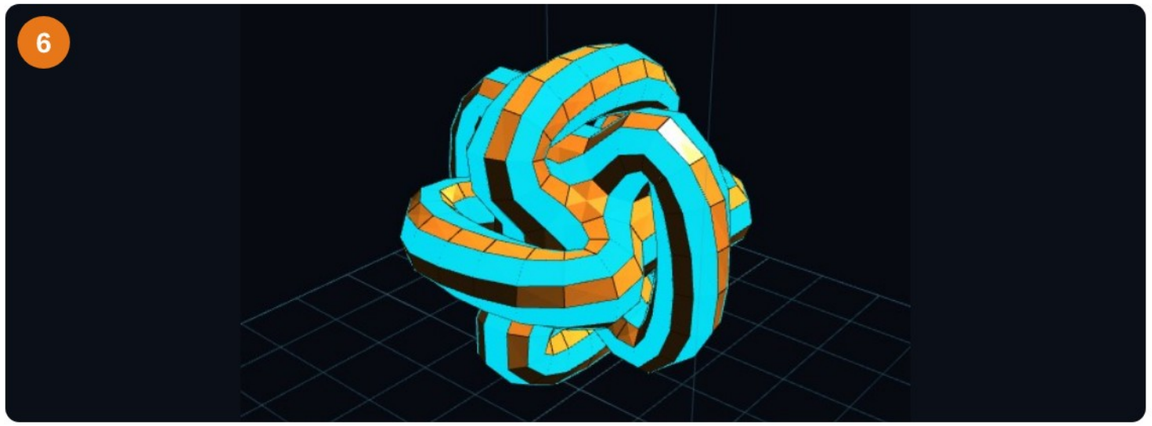
DOO-SABIN SUBDIVISION

$v' = \frac{1}{4}\,(v + c_f + m_- + m_+)$

c_f = face centroid · $m_\pm$ = corner edge midpoints

faces: $F' = F + E + V$

V 720 − E 1440 + F 710 = χ -10 ⇒ genus 6



Select face loops

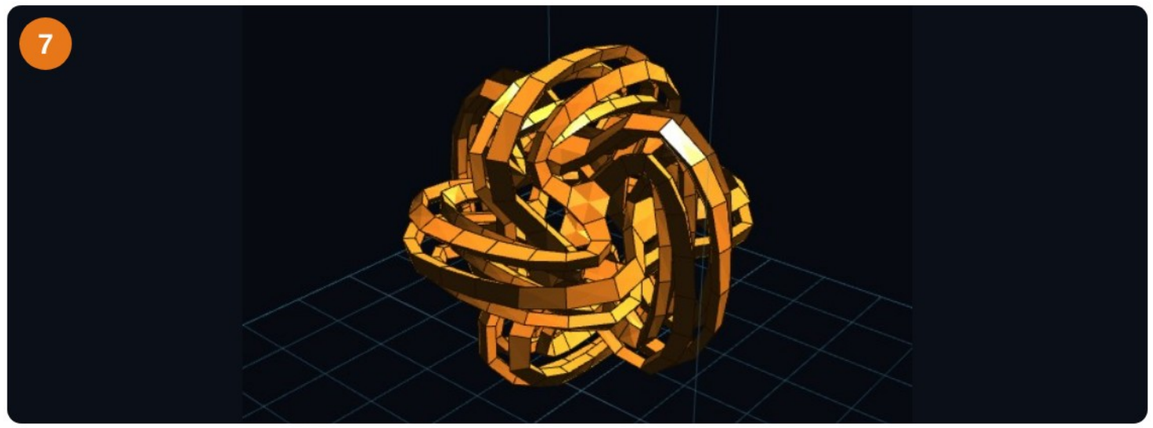
Tools ▸ Loops. Click faces along the handles — the loop runs across the edge nearest your click. Choose 3 loops that never touch.

FACE LOOP (TOPMOD “SELECT FACE LOOP”)

walk quad → opposite edge → next quad …

stop at a non-quad or back at the start

V 720 − E 1440 + F 710 = χ -10 ⇒ genus 6



Rind modeling

Tools ▸ Rind. The surface is thickened into a crust and the selected loops are cut out as slots — walls are built only along their outline.

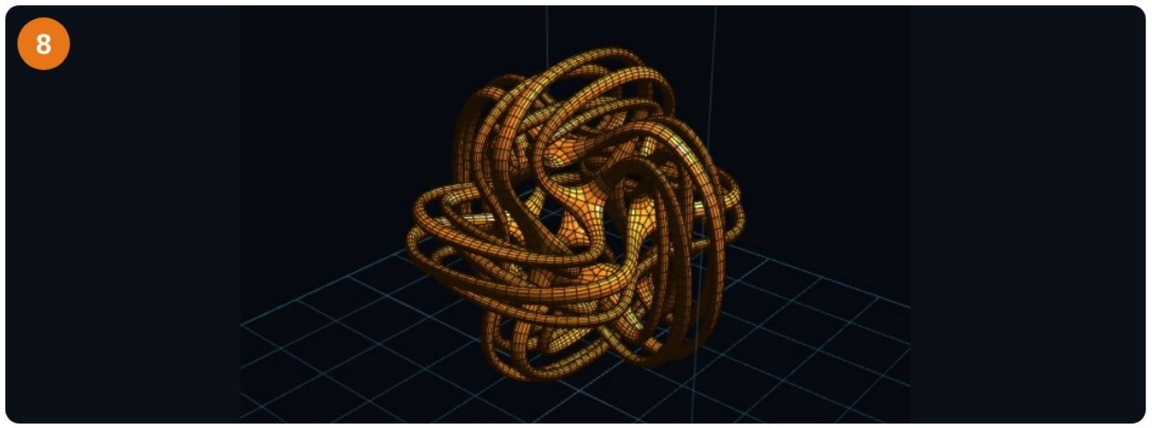
RIND MODELING

inner shell $v_{in} = v - t \cdot \hat{n}_v$

remove face set S from both shells

walls on ∂S : [a, b, b_{in}, a_{in}]

V 1440 − E 2880 + F 1420 = χ -20 ⇒ genus 11



Catmull-Clark ×2

Remeshing ▸ Catmull ×2 — smooths every face; the slots stay open.

CATMULL-CLARK SUBDIVISION

face $f = (1/n)\,\Sigma\, v_i$

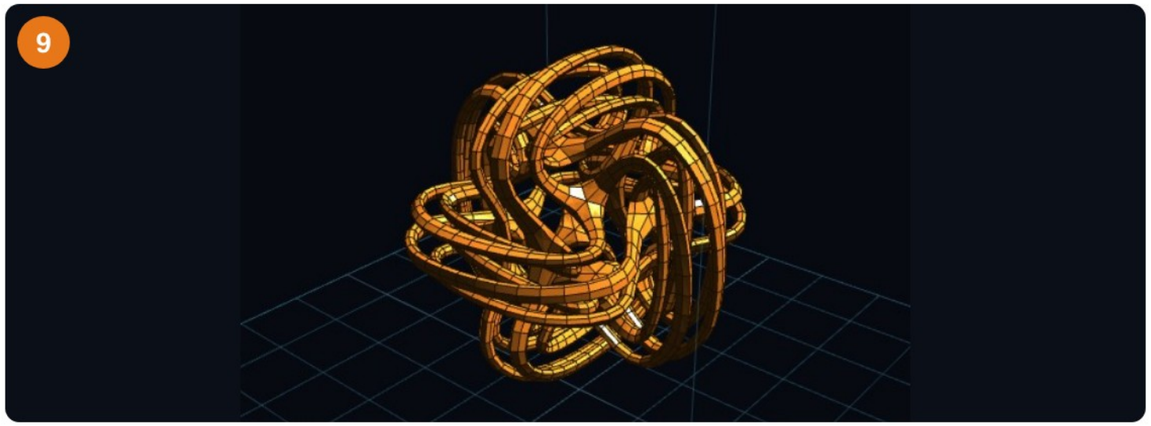
edge $e = (v_0 + v_1 + f_0 + f_1) / 4$

vertex $v' = (F + 2R + (n - 3)\cdot v) / n$

V 23020 − E 46080 + F 23040 = χ -20 ⇒ genus 11

PROJECT	SHEET TITLE	SOURCE	DRAWN BY	DATE	STEPS	SHEET
The TopMod Look	**STEPS 5–8**	TOPMOD 3D v3.9	TOPMOD 3D	2026-10-13	11	4 OF 6 **04**

9



Undo one level

⌘Z once to step back to a single level of subdivision — lighter mesh, same form.

CATMULL-CLARK SUBDIVISION

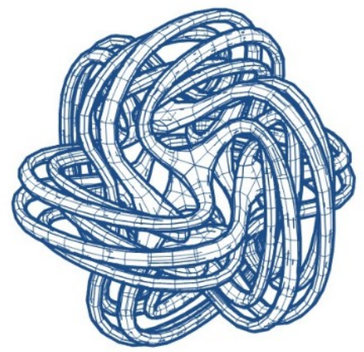
face $f = (1/n)\,\Sigma\, v_i$

edge $e = (v_0 + v_1 + f_0 + f_1) / 4$

vertex $v' = (F + 2R + (n - 3)\cdot v) / n$

V 5740 − E 11520 + F 5760 = χ -20 ⇒ genus 11

10



Line Art

View style ▾ Line Art — wireframe with hidden lines removed, blue on white. Export SVG / PNG / HPGL from the Options window.

LINE ART (PLOTTER)

silhouette: $(\hat{n}_0\cdot v)(\hat{n}_1\cdot v) < 0$

crease: $\hat{n}_0\cdot\hat{n}_1 < \cos\theta$

hatch where $½ + ½\,\hat{n}\cdot\hat{l} <$ tone

V 5740 − E 11520 + F 5760 = χ -20 ⇒ genus 11

11



Sapphire view

View style ▾ Sapphire. Ray-traced refraction, total internal reflection and Beer absorption; 💡 sets the light.

RAY-TRACED GEM

Snell $n_1 \sin\theta_1 = n_2 \sin\theta_2$

Fresnel $R = ½\,(r_s^2 + r_p^2)$

total internal reflection $\theta > \theta_c = \mathrm{asin}(1/n)$

V 5740 − E 11520 + F 5760 = χ -20 ⇒ genus 11

PROJECT	SHEET TITLE	SOURCE	DRAWN BY	DATE	STEPS	SHEET
The TopMod Look	**STEPS 9–11**	TOPMOD 3D v3.9	TOPMOD 3D	2026-10-13	11	5 OF 6 **05**

Step reference

#	STEP	MATH	V	E	F	GENUS
1	Dodecahedron	12 pentagons · 30 edges · 20 vertices	20	30	12	0
2	Handle 1 · 72° twist	each handle: $\chi \to \chi - 2 \cdot g \to g + 1$	45	85	40	1
3	Handles 2–3	each handle: $\chi \to \chi - 2 \cdot g \to g + 1$	95	195	96	3
4	All 6 handles	each handle: $\chi \to \chi - 2 \cdot g \to g + 1$	170	360	180	6
5	Doo-Sabin ×1	$v' = \frac{1}{4}\,(v + c_f + m_- + m_*)$	720	1440	710	6
6	Select face loops	walk quad → opposite edge → next quad …	720	1440	710	6
7	Rind modeling	inner shell $v_in = v - t \cdot \hat{n}_v$	1440	2880	1420	11
8	Catmull-Clark ×2	face $f = (1/n)\,\Sigma\, v_i$	23020	46080	23040	11
9	Undo one level	face $f = (1/n)\,\Sigma\, v_i$	5740	11520	5760	11
10	Line Art	silhouette: $(\hat{n}_0 \cdot v)(\hat{n}_1 \cdot v) < 0$	5740	11520	5760	11
11	Sapphire view	Snell $n_1 \sin\theta_1 = n_2 \sin\theta_2$	5740	11520	5760	11

PROJECT	SHEET TITLE	SOURCE	DRAWN BY	DATE	STEPS	SHEET
The TopMod Look	**REFERENCE**	TOPMOD 3D v3.9	TOPMOD 3D	2026-10-13	11	6 OF 6 **06**